LINUX BASICS

Quick-reference cheat sheet for UVM verification students

How to read examples $ is the terminal prompt—do not type it. Replace text inside <angle brackets> with your own value. Linux is case-sensitive.

1 Navigate

Command	What it does / example
`$ pwd`	Show the full path of the current directory.
`$ ls`	List files and directories.
`$ ls -la`	Detailed list, including hidden files.
`$ cd <directory>`	Enter a directory. Example: cd project
`$ cd ..`	Move up one directory.
`$ cd ~`	Go to your home directory.

2 Files and directories

Command	What it does / example
`$ mkdir <name>`	Create a directory.
`$ touch <file>`	Create an empty file.
`$ cp <source> <destination>`	Copy a file. Add -r to copy a directory.
`$ mv <source> <destination>`	Move or rename a file/directory.
`$ rm <file>`	Permanently delete a file.
`$ rm -r <directory>`	Permanently delete a directory and its contents.

3 View and search text

Command	What it does / example
`$ cat <file>`	Print the entire file.
`$ less <file>`	Open a scrollable view; press q to quit.
`$ head <file>`	Show the first 10 lines.
`$ tail <file>`	Show the last 10 lines.
`$ tail -f <logfile>`	Follow new log output; press Ctrl+C to stop.
`$ grep "text" <file>`	Show lines containing text.
`$ grep -in "error" <file>`	Search ignoring case (-i) and show line numbers (-n).
`$ grep -R "text" <directory>`	Search recursively through a directory.

CAUTION rm bypasses the recycle bin. Check the path before pressing Enter; avoid rm -rf unless you fully understand it.

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4 Combine commands and save output

Command	What it does / example
`$ command > output.log`	Write output to a file, replacing existing content.
`$ command >> output.log`	Append output to a file.
`$ command 2> errors.log`	Write error messages to a file.
`$ command | less`	Send output into a scrollable viewer.
`$ command | grep "error"`	Keep only output lines containing error.
`$ command1 && command2`	Run command2 only if command1 succeeds.

5 Processes and jobs

Command	What it does / example
`$ ps -ef`	List running processes.
`$ top`	Monitor processes; press q to quit.
`$ command &`	Run a command in the background.
`$ jobs`	List jobs started from this shell.
`$ fg`	Bring the most recent job to the foreground.
`$ kill <PID>`	Request that a process stop.
`Ctrl+C`	Stop the command currently running in the foreground.

6 Useful essentials

Command	What it does / example
`$ chmod u+x <file>`	Allow yourself to execute a script or program.
`$ ./<program>`	Run a program from the current directory.
`$ history`	Show recent commands.
`$ <command> --help`	Show short help for a command.
`$ man <command>`	Open the manual; press q to quit.
`$ echo $<VARIABLE>`	Print an environment variable. Example: echo $PATH
`$ source <script>`	Run a setup script in the current shell.

7 Paths and keyboard shortcuts

Command	What it does / example
`~`	Your home directory.
`.`	The current directory.
`..`	The parent directory.
`Tab`	Complete a command or filename.
`↑ / ↓`	Move through command history.
`Ctrl+L`	Clear the terminal screen.